E-Waste Regulations Comparative Analysis: India vs Global Frameworks

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E-Waste Management Rules

A Comparative Study: India, United States, Australia, Russia & Japan

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1. Introduction

Electronic waste (e-waste) is one of the fastest-growing waste streams in the world, driven by shrinking product lifecycles and rising consumption of electrical and electronic equipment (EEE). Countries have responded with very different regulatory philosophies — from India's centralized, portal-driven Extended Producer Responsibility (EPR) system, to the fragmented state-by-state approach in the United States, to Japan's decades-old dual-law model. This document compares the e-waste management frameworks of India, the United States, Australia, Russia, and Japan across governance, responsibility, targets, and enforcement.

2. Country-by-Country Comparison

Parameter

India

United States

Australia

Russia

Japan

Governing Framework

E-Waste (Management) Rules, 2022

(in force since 1 Apr 2023), replacing the 2016 Rules; amended 2023 & 2024

No comprehensive federal law;

a patchwork of ~25-26 state laws (e.g. California, New York, Illinois)

Recycling and Waste Reduction Act 2020

+ Product Stewardship (TVs & Computers) Rules 2021

Federal Law No. 89-FZ on Production

and Consumption Waste, plus EPR decrees; e-waste largely folded into general/hazardous waste rules

Home Appliance Recycling Law (2001)

+ Small Home Appliance Recycling Act (2013) + Law for Promotion of Effective Utilization of Resources

Regulatory Body

Central Pollution Control Board (CPCB)

under Ministry of Environment, Forest & Climate Change

US EPA (limited, hazardous-waste focus);

individual state environmental agencies

Dept. of Climate Change, Energy,

Environment & Water (DCCEEW); co-regulatory bodies

Rosprirodnadzor

(Federal Service for Supervision of Natural Resources)

Ministry of Environment &

Ministry of Economy, Trade and Industry (METI)

Core Policy Model

Extended Producer Responsibility (EPR) with mandatory online recycling targets

Mixed models: state-level EPR, fee-based, or advance recycling fee, depending on state

Co-regulatory / industry-funded product stewardship scheme

EPR-based eco-fee system; e-waste covered indirectly within general producer-responsibility law

Manufacturer-funded EPR with statutory consumer recycling fees for large appliances

Who Pays / Is Responsible

Producers (manufacturers, importers, brand owners) fund and execute recycling via registered recyclers

Varies by state — often producers, sometimes retailers or consumers via fees

Importers/manufacturers of TVs & computers ("liable parties") fund collection & recycling

Manufacturers/importers pay eco-fee or self-arrange disposal per recycling standards

Consumers pay a recycling fee at disposal (4 major appliances); producers fund small-appliance recycling

Coverage / Scope

106 EEE items across 7 categories, incl. solar PV modules (special storage rules to 2034-35)

Varies widely: monitors, computers, TVs, cell phones; some states cover very few devices

Focused mainly on TVs and computers/peripherals; other e-waste left to states/territories

Falls under general "finished goods to be recycled" list, incl. electronics; scope less item-specific

4 large appliances (TV, fridge, washing machine, A/C) + broad small-electronics category

Recycling Targets

Phased targets: 60% (FY23-24, FY24-25) → 70% (FY25-26, FY26-27) → 80% (FY27-28 onward)

No national target; state targets vary (where they exist at all)

Rising targets under NTCRS, moving toward 80% by 2026-27 for covered products

Recycling-rate targets set by government decree, sector-specific eco-standards

No fixed national %; law mandates collection/handling systems, with steadily rising item-level recovery

Compliance Mechanism

Digital EPR portal; tradable EPR certificates; environmental compensation for shortfalls

State registration/reporting; certification via R2 or e-Stewards recyclers (voluntary but market-standard)

Reporting to DCCEEW by co-regulatory arrangements on behalf of liable parties

Unified state waste registration system; automatic eco-fee calculation by Rosprirodnadzor

Retailer take-back obligation; recycling fee tickets (kaden recycling ken) tracked through the system

Informal Sector

Large informal recycling sector; rules aim to formalize it but integration remains a challenge

Minimal informal sector; issue is more regulatory fragmentation than informality

Small informal sector; landfill diversion/illegal dumping in bins is the bigger issue

Informal handling still common outside registered regional operators

Very limited informal sector; highly formalized retailer/manufacturer collection network

Notable Recent Development

2024 amendment created a trading platform for EPR certificates; solar e-waste rules added

US now enforces Basel Convention import controls (from Jan 2025) even though it hasn't ratified Basel itself

Industry (ACOR) is pushing for a broader national EPR scheme beyond TVs/computers

From Jan 2025, EPR obligations broadened to cover 100% of packaging waste recycling, tightening the wider producer-responsibility regime

One of the earliest EPR pioneers (since 2001); system now a global reference model

3. Key Observations

3.1 EPR Is the Common Thread, but Implementation Differs Sharply

India runs one of the world's most centralized EPR systems — a single national digital portal with tradable EPR certificates and phased recycling targets rising to 80% by FY 2027-28.

Japan pioneered EPR as early as 2001, but splits responsibility: consumers pay recycling fees on four major appliances, while producers fund and manage collection of smaller electronics.

Australia's EPR is narrower in scope, focused mainly on TVs and computers through the co-regulatory NTCRS, leaving other e-waste categories to state and territory governments.

The United States has no unified EPR law; roughly half its states have some form of producer-responsibility legislation, creating compliance complexity for multi-state operators.

Russia's EPR regime is younger and broader in ambition (recently expanded to 100% packaging-recycling obligations from January 2025), but e-waste itself is not treated as a distinct, heavily itemized category the way it is in India or Japan.

3.2 Regulatory Maturity and Enforcement

India and Japan have the most detailed, item-specific rules with clear numeric targets and designated regulators (CPCB and METI/Ministry of Environment respectively).

Australia and the U.S. rely more heavily on industry co-regulation or state discretion, resulting in more variable enforcement and transparency.

Russia's system is still consolidating — a unified waste-registration system and automatic eco-fee calculation by Rosprirodnadzor are relatively recent additions.

3.3 Informal Sector Challenges

India faces the largest informal-sector challenge among the five, given the scale of its e-waste generation and the entrenched informal recycling economy; its 2022 Rules explicitly aim to channel more volume into the formal, registered-recycler system. Japan and Australia, by contrast, operate largely formal collection networks, while the U.S. and Russia sit in between, with fragmentation and inconsistent enforcement being bigger issues than informal recycling itself.

4. Summary Table — At a Glance

Feature

India

United States

Australia

Russia

Japan

National EPR law?

Yes

No (state-level only)

Partial (TVs/computers)

Yes (broadening)

Yes

Central digital compliance portal?

Yes

No

No (per co-regulator)

Emerging

Partial

Numeric recycling targets?

Yes (phased to 80%)

Rare / state-dependent

Yes (to 80% by 2026-27)

Yes (sector decrees)

No fixed national %

Who pays at end-of-life?

Producer

Varies

Producer/importer

Producer (eco-fee)

Consumer (large items)

Sources: CPCB E-Waste (Management) Rules 2022 & FAQs; DCCEEW (Australia) NTCRS documentation; Japanese Law Translation database; US EPA and state-law compliance summaries; Rosprirodnadzor / Russian federal waste-law commentary. Figures reflect publicly available regulatory information as of August 2026 and may be updated by ongoing amendments in each jurisdiction.